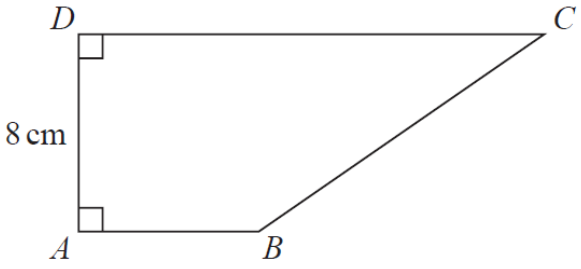
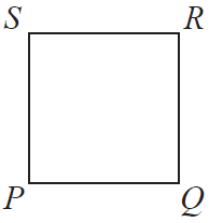
**26** The diagram shows a trapezium *ABCD* and a square *PQRS*.




$AD = 8$ cm
$DC = 3AB$

The perimeter of the square is 24 cm.

The area of the square is half the area of the trapezium.

Work out the length of $AB$.

Square side = 24 ÷ 4 = 6 cm

**1 mark**
Find the square area.

Square area = $6^2$ = 36 $cm^2$

Trapezium area = 2 × 36 = 72 $cm^2$

Let AB = x, so DC = 3x.

**1 mark**
Use trapezium area formula.

A = ½(a + b)h: 72 = ½(x + 3x) × 8

**1 mark**
Set area equal to 72.

72 = 16x, so x = 72 ÷ 16 = 4.5 cm

**1 mark**
AB = 4.5 cm.

..... cm

**(Total for Question 26 is 4 marks)**